Quantum Computing Fundamentals

Cornell Quantum Computing Association

Fall 2025 — Saturdays 2:00-3:00 PM — Rockefeller 122
Instructor: Arnab Ghosh — **Discord:** `https://discord.gg/FXPXeqgDm6`

This program provides a **comprehensive introduction to quantum computing** for students with **minimal prior knowledge**. We build from **basic quantum mechanics principles** to **real-world hardware implementations**, emphasizing **hands-on experience** through interactive programming exercises. Students will **understand fundamental quantum concepts**, **implement basic quantum algorithms**, and gain **practical experience with quantum computing platforms**.

Format: Each 60-minute session includes 5 minutes reviewing the previous week's worksheet and 55 minutes of interactive lecture with live demonstrations. Materials include lecture slides, supplementary worksheets, Classiq notebooks for visual circuit design, and select Qiskit notebooks for advanced implementations and real hardware access. All platforms are free and web-based. Course announcements and discussions occur on the QCA Discord server.

Course Schedule

Week 1 — September 27 **Introduction to Quantum Computing**
Classical vs quantum paradigms, qubits, Bloch sphere visualization, basic measurement operations

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Week 2 — October 4 **Superposition & Quantum Gates**
Mathematical representation of quantum states, Hadamard gate, quantum circuit notation

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October 11 **No Class - Fall Break**

Week 3 — October 18 **Entanglement**
Bell states and EPR pairs, two-qubit gates, quantum correlations and non-locality

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Week 4 — October 25 **Deutsch's Algorithm**
First quantum advantage demonstration, oracle implementation, circuit construction

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Week 5 — November 1 **Quantum Teleportation**
No-cloning theorem, quantum communication protocols, Bell measurement

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Week 6 — November 8 **Advanced Algorithms Survey**
Shor's factorization, Grover's search, Quantum Fourier Transform, cryptography

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Week 7 — November 15 **Quantum Hardware**
Qubit technologies, current system performance metrics and limitations

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November 29 **No Class - Thanksgiving Break**

Week 8 — December 6 **Error Correction & Future**
Fault tolerance requirements, NISQ era limitations, quantum advantage timeline